

# A Mixed-Methods Analysis of Account Creation & Authentication Inaccessibility for Blind and Low Vision Users

Adryana Hutchinson  
adryana.hutchinson@gwu.edu  
The George Washington University  
Washington, D.C., USA

Matthias Fassl  
matthias.fassl@gwu.edu  
The George Washington University  
Washington, D.C., USA

Smirity Kaushik  
smirity.kaushik@gwu.edu  
The George Washington University  
Washington, D.C., USA

Adam J. Aviv  
aaviv@gwu.edu  
The George Washington University  
Washington, D.C., USA

## ABSTRACT

For many blind and low-vision (BLV) users, interacting with security and privacy mechanisms, such as password policies and two-factor authentication, presents significant accessibility barriers. Likewise, interacting with account creation (sign-up) and authentication (login) mechanisms, such as email, password, and other fields/elements on sign-up and login pages, can also present serious barriers to BLV users. While prior work has highlighted the inaccessibility of specific mechanisms, understanding how the broader inaccessibility of elements on account creation and authentication interfaces impacts both security and privacy decisions remains underexplored. To assess the breadth and depth of these challenges, we conducted a two-part, mixed-method study, first using automated auditing to measure accessibility issues across 11,273 sign-up and login pages (including 325 manual audits). We then conducted 14 semi-structured interviews to explore BLV users' broader experiences. Through the measurement, 81% of the identified errors pose potentially significant barriers to people with disabilities. Expanding on those findings through interviews, we further identified seven major inaccessibility categories, including inoperable buttons, unlabeled fields, and inaccessible error messages. Participants also reported significant friction with other mechanisms, such as CAPTCHAs and recovery mechanisms, electing not to use these systems to avoid inaccessibility fatigue. Our findings highlight a disconnect between current security and privacy practices and the needs of BLV users. We provide specific recommendations for practitioners to improve authentication accessibility holistically.

## CCS CONCEPTS

- **Security and privacy** → **Usability in security and privacy**;
- **Human-centered computing** → **User studies**; **Accessibility design and evaluation methods**.

## KEYWORDS

Security, Authentication, Privacy, Accessibility, Blind and Low Vision Users, Mixed-Methods

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## 1 INTRODUCTION

Blind and low vision (BLV) users are likely to encounter significant accessibility barriers that hinder their experiences on the web, with recent accessibility reports indicating that over 95% of the Internet is inaccessible to people with disabilities [82]. Part of this inaccessibility stems from web formatting that ignores how screen readers read through pages. For BLV users who use screen readers, navigating complex interfaces while concurrently processing auditory information makes it especially difficult to complete web-based tasks in a timely manner [28]. This extends to the most basic processes that sighted users take for granted: sign-up (account creation) and login (authentication) forms.<sup>1</sup> These interfaces require that users make decisions that impact their security and privacy: choosing what username, password, 2FA mechanism, or what information to withhold from a website, are all choices that can determine a user's level of vulnerability to compromise.

The usable security and privacy community has long focused on developing interventions that empower users to engage with mechanisms that facilitate meaningful decision-making, such as privacy labels [39], nudges [1], dashboards [26], and password strength meters [73]. Prior research has also highlighted that BLV users experience significant obstacles when using security and privacy mechanisms [22–24, 48, 58], including the inability to navigate cookie banners [18], enable two-factor authentication [22–24, 48, 58], or see web security alerts [53]. All of the aforementioned processes require that users make rapid security and privacy decisions which, depending on design decisions, can help or hinder them in making

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<sup>1</sup>We define sign-up forms as account creation interfaces (login info, personal info, password policies and other fields) and login forms as authentication mechanisms (credentials, 2FA, and password recovery). Throughout this paper, we refer to both collectively as sign-up and login forms.

safe decisions [19, 34, 62, 73]. When BLV users are forced to split their cognitive resources between navigation and auditory output from their assistive technology, it can also hinder their ability to make these decisions. In this paper, we make the case that this also applies to sign-up and login forms. While prior work has identified significant accessibility barriers in specific aspects of these processes (e.g., 2FA [4], password management [61]), a large-scale analysis of the inaccessibility of account creation and authentication, and the resulting security and privacy harms, has yet to be conducted. Furthermore, work that focuses on the web authentication experiences of BLV users is over a decade old [23], a period during which the web landscape has evolved drastically [3, 43, 60].

To bridge this gap, our work aims to assess both the breadth and broader impact of accessibility violations during account creation (sign-up) and authentication (login), and how they may play a role in how users make security and privacy decisions. We ask the following research questions:

- (RQ1) How often do accessibility violations appear during account creation and authentication, and what are the most common interaction-related accessibility issues?
- (RQ2) What are the security and privacy implications of unaddressed accessibility violations during account creation and authentication?
- (RQ3) What are blind and low vision users' experiences with account creation and authentication?

To address RQ1 and RQ2, we measured ( $n = 11,273$ ) sign-up and login forms using an established automated accessibility audit framework [20]. To find accessibility errors not covered by the automated audits, we also performed manual audits on 350 sign-up and login forms, which involves manually interacting with each page element using a screen reader. We found that the website corpus had a total of 28,248 **critical** accessibility issues, including inaccessible buttons, improperly labeled input fields, and inaccessible error messages. These issues have a substantial impact on users, especially those with disabilities that rely on screen readers [24], such as preventing access to error messages or not specifying where a link takes them. Fortunately, many of these accessibility issues can be mitigated by simple fixes, such as labeling input fields appropriately and using accessible HTML form indicators. More complex issues, such as meaningful webpage formatting or assistive technology support, may need other design considerations, which we cover in the Discussion 6 section.

To further explore the depth of experiences during account creation and authentication, we conducted ( $n = 14$ ) semi-structured interviews with BLV users (RQ3). We find that a majority of our participants have experienced some form of inaccessibility that can hinder their security and privacy decisions during account creation or authentication. Most participants, for example, cannot discern the error messages on sign-up or login forms, or do not enable two-factor authentication unless forced to due to setup difficulty and system-specific inaccessibility, and incompatibility with their screen readers. We also found nuanced issues not covered by web audits, such as password recovery difficulties and navigation strategies, such as double-checking input fields and pages for errors. Some of the issues we identified, such as password policy inaccessibility, have also not been investigated in prior work.

Through our manual audits and qualitative results, we underline how specific inaccessible elements affect security and privacy. Likewise, while an inaccessible authentication interface does not definitively mean a BLV user will be forced to make an insecure decision, we also argue that BLV users who use assistive technology will likely to experience choice paralysis and information fatigue caused by how information is presented on these interfaces [28]. As usable security research has highlighted, and some of our participants have noted, this can result in insecure behaviors [19, 31].

## 2 BACKGROUND AND RELATED WORK

**WCAG Guidelines.** The WCAG are a set of globally recognized guidelines that inform how web content should be made accessible to people with disabilities [79]. The guidelines are built on four main principles, which we summarize below:

- **Perceivability:** This guideline pertains to content presentation. To comply, non-text elements like images and graphics must include alt-text, while video content requires captioning.
- **Operability:** To be operable, web content and navigation must be keyboard-accessible and logically structured.
- **Understandability:** Broadly, this guideline aims to ensure that information is understandable to all audiences.
- **Robustness:** Robustness refers to content being able to be accessed by a variety of devices, such as assistive technology.

Additionally, the WCAG categorizes success criteria into three levels of increasing stringency: A (minimum), AA (ideal), and AAA (maximum). Level A is the baseline, and violating its standards creates significant and severe barriers for users [21]. Level AA is the target standard for most organizations, including US government websites, as it addresses the needs of the majority of people with disabilities [84].

**Web Authentication Usability.** Users' web authentication experiences have seen exhaustive research within the security and privacy space. In order to cope with the growing complexity of authentication schemes, users often adopt unsafe security practices such as choosing memorable – but not necessarily secure – passwords [27]. Many factors play a role in password choice, including account importance [55, 74], expected use cases (such as using the account on a mobile device) [77], and the policies of specific websites [41, 66]. In lieu of insecure practices, security and privacy practitioners have developed better security mechanisms to ensure users protect their accounts, including password policies [41, 66] and strength meters [73], password managers [56, 59], and two-factor authentication (2FA) mechanisms. Despite these developments, users still struggle with creating, remembering, and changing their passwords for every website due to the heavy mental strain. As a consequence, users will often choose to reuse their passwords [74]. Even when password policies are enforced, users will still reuse most of their password, and only make necessary character changes to meet policy requirements [66].

2FA mechanisms can also pose an issue to usability. Users will reject mechanisms that are opaque, cumbersome, or insecure [50]. Particularly relevant are issues with authenticator applications – able users have reported that both the setup process [63] and general use these applications can be unusable [50]. Despite users

often practice unsafe web authentication habits, properly configured password policies can reduce risk. For example, researchers have found that password policies with password blacklists generate more secure passwords from users [70]. This assumes that information about the password is properly conveyed to the user; unfortunately, we find that many websites in our dataset do not generate accessible warnings or suggestions.

**Web Accessibility & Measurement.** A number of web measurement analyses inform this work. Alroomi and Lee [6], for example, measured the password policies of over 20k websites to develop a taxonomy of popular password policies. Another study by Yuqi et al. [34] surveyed the implementations of password masking and behavior on 66 sites, noting that over half do not allow for password mask toggling. This can present an accessibility issue to users. Web accessibility measurement, and remediation, is also an area of concern. In 2010, Lopes et al. analyzed over 28 million webpages, and found poor compliance levels among the general web ecosystem [46]. Importantly, prior work has also found that most BLV users do not use ad blockers to block these ads [87], which makes them particularly vulnerable to malicious pop-ups during account creation and login. While some technologies, such as accessibility overlays, promise to mediate some of these issues, they rarely do; a study by Makati et al. [49] found that these tools actually made accessibility worse for BLV users.

**Accessible Authentication.** Authentication has presented numerous problems for people with disabilities. In 2015, Dosono et al. qualitatively analyzed BLV participants' experience with web authentication, finding that users often use context clues to understand what page they are on when authenticating – as the researchers note, users must overcome a monumental number of accessibility violations in order to successfully log into accounts [23]. Similarly, Brenton and AbdelRahman [16] developed a framework that measures screen reader accessibility violations on landing pages, and how privacy threats may manifest from those violations. We find that our participants share similar frustrations with being unable to know what they are entering or how their information is used. Furthermore, we measure different violation avenues and assess broader accessibility violations, allowing us to capture a more nuanced set of disability-related issues and experiences.

Other concerns, such as 2FA [4] and mobile phone authentication [17] have also presented accessibility issues to BLV users, which feature heavily in our own interviews. A recent SOK paper by Akanda et al. analyzed novel authentication solutions, specifically designed for BLV users, and found that many of the mechanisms were both insecure and failed to meet the accessibility needs of the users they were designed for [3]. This suggests a divide between theoretical design principles and the practical security and accessibility needs of disabled users, even with technologies specifically designed for them. Our work further confirms this in regards to account creation and authentication more broadly.

Password managers are also used by BLV users. Ponticello et al. found that adoption was widely driven by convenience, with many users finding the autofill features of password managers especially useful [61]. Autofill removes a cumbersome part of the authentication process: entering information into forms. However, as the authors note, users do not use other password manager features

(strong password generation) due to the inaccessibility of the interfaces themselves. This compliments our findings by highlighting how overarching structural inaccessibility impacts overall security decisions, such as using a strong password generator.

Our work is greatly inspired by Dosono et al. [23]; notably, we find that many of the issues outlined in their work are **still** prevalent over a decade later on many websites. However, we also find many novel issues not explored in their study, such as issues related to password policies, strength meters, and autofill. Further, our work measures the prevalence of these issues quantitatively, which is relatively lacking in accessible S&P research. This allows us to illustrate the sheer amount of issues on authentication webpages where inaccessible interactions, whether due to inaccessible elements or assistive technology incompatibility, risks compounding cognitive load for BLV users in information-rich scenarios [28]. Allowing users to make informed decisions about their security and privacy has been a long-standing research objective in usable S&P [1, 39, 73] yet understanding how accessibility practices deteriorate this ability has scanty been explored within the space. We seek to rectify this with this work.

## 3 METHODOLOGY

### 3.1 Quantitative Website Accessibility Audit

In this section, we outline the methodology for analyzing and categorizing websites in our corpus. Notably, during our analysis, we only considered Level A and AA accessibility violations. While Level AAA and best practice violations matter, our immediate focus is on achieving broader adoption of Level A and AA guidelines, which have not yet been universally adopted even among the general web [82]. We discuss Level AAA and best practice compliance measurement in our Discussion 6. We used a Windows system and Chrome browser during our web measurement. We first conducted the automated accessibility assessment, followed by the manual audit.

**Automated High-Level Audits.** There were two rounds of automated accessibility assessments on our corpus. The first audit focused on identifying initial, automatically detectable accessibility issues, such as missing alternative text or low-contrast elements. We then filtered the dataset to identify any false positives. Our website corpus was generously provided to us by Alroomi and Li, who initially gathered sign-up links to analyze password policies at scale [6], collecting ( $n = 20,120$ ) total webpages in 2023. We began our first audit by using Puppeteer [71] and Deque's axe-core [20] accessibility tools, which generate JSON-formatted accessibility audits of webpages. Axe-core identifies issues that can be automatically detected, such as low-contrast, invalid alt text, and duplicate form names. This initial audit took place between November 2024 and January 2025.

Upon manual inspection of the websites within our dataset, we found false positives – specifically, websites that were neither login nor sign-up forms, but rather initial landing pages (or pages that no longer exist). To ensure data quality, we used Senol et al.'s [68] authentication page classifier extension to filter pages in our corpus as login, sign-up, or neither. Pages that were neither were discarded from analysis. This took place in March 2025. 8,847 websites were

invalid. During filtering we saw that the tool removed the found false positives in the initial search. For further confirmation, the first author then manually looked at a random selection of 100 websites, and found them to all be either sign-up or login forms. To further elucidate potential inaccessibility patterns in our dataset, we also performed an analysis on the distribution of accessibility violations based on website category using Wettig et al.'s [83] website classifier. Since the classifier classifies a website based on page text, we used the base domain of each website to categorize them.

**Manual In-Depth Audits.** While automated audits are able to capture technical issues, they are rarely able to highlight nuanced contextual violations, such as lack of information availability or improper form structure. Standard accessibility practices include both automated and manual approaches [67], which we adopt for this study. As such, for our manual analysis, we selected a representative sample of both popular and less popular websites from the dataset using their Tranco rank [44], a measure of website traffic. We used the Tranco list that collected its data from February 2025 to March 2025. We then took an even selection of popular (above the median Tranco rank within our dataset) and less popular websites, with the most popular website having a Tranco rank of 11, and the least popular having a rank of 4,247,512. In total, a subset of 325 (35 login and 290 sign-up) authentication pages were selected for in-depth manual audits, which were selected randomly.

The WCAG comprises a total of 13 guidelines, each with a corresponding set of 87 success criteria [79]. Because many of the success criteria do not pertain to user interaction with web pages or form fields (see Section 2), we limited the scope to measuring only interaction-related accessibility guidelines outlined in the Interaction Design subsection [38] of the WCAG. If the auditor noticed a non-interaction-related issue that directly impacted the accessibility of the account creation form, they were instructed to include that issue in their report. We used NVDA throughout our analysis, an open-source screen reader that's second in popularity only to JAWS [81]. The first author regularly uses NVDA for web browsing.

Additionally, the first author, who has industry-level accessibility experience, performed all the manual audits. Throughout the manual review process, the auditor used a custom web application to manually enter the names and amounts of applicable violations. The auditing procedure was as follows:

- Load the website and wait 15 seconds before proceeding to determine if any pop-ups capture the user's mouse position without notifying them with a pop-up tag or alert.
- Starting with the first selectable element, process through the webpage to identify issues with tabbing or element outlines.
- Check that the form fields have appropriate tags, labels, and error messages. Error messages were tested by first only entering an invalid email ('@gmail.com') and clicking the confirmation button. With the email field cleared, a dummy password ('a') was then entered before clicking the confirmation button again.
- Check that all interactable elements on the page are properly labeled and accessible.
- Finally, check when a password policy is visible by entering the dummy password and clicking the confirmation button.

**Password Policies.** Furthermore, the auditor was also tasked with identifying when, if any, password policies (PWP) appeared to the user. This was done to understand how a BLV user may interact with the PWP, which we ask about during our interviews. The auditor indicated that a website had either:

- **Accessible PWP:** The policy is visible before or during password entry (47 total).
- **Inaccessible PWP:** The form field indicates that an error had occurred, but no policy is displayed (63 total).
- **Delayed PWP:** The policy only appears after invalid information is entered and the confirmation button is pressed (133 total).
- **No PWP:** Our dummy password of 'a' was accepted (82 total).

### 3.2 Interview Study

To understand the implied security and privacy harms of inaccessible account creation and authentication, we conducted semi-structured interviews with ( $n = 14$ ) BLV participants. The interview protocol was informed by the harms the auditor cataloged in the manual audit, taking care to ask about experiences with password policies and password meters that were rarely accessible.

**Interview Protocol.** Interviews took place over Zoom and lasted an average of 40 minutes. Before starting each interview, we obtained participants' verbal consent to participate and to be audio recorded. Those who did not want to be recorded were transcribed using Zoom's auto-transcription feature. We also informed them about our study procedure and data protection policy.

We developed the preliminary interview protocol based on the accessibility issues identified in the quantitative audit of the webpages. For example, we identified issues related to unlabeled check boxes, pop-ups, and password masking, among others. Upon piloting our study with lab members and close contacts, we refined the preliminary interview protocol by cutting down on redundancies, rephrasing questions, and including more probing questions for technical issues, such as password policies or recovery. We began by asking participants to provide their demographic details, such as age, gender, education, and whether they attended a school for the blind, followed by information on their technology use. We then probed for concerns related to credential creation, form field navigation, and overall security and privacy behavior more broadly, such as password and account creation strategies. This allowed accessibility concerns that were not originally considered by the researchers to naturally emerge during conversations. At the end of each interview subsection, we asked about specific problems present in our dataset if they were not brought up organically. For example, when discussing the use of autofill, we concluded with a question about any security concerns related to its use. We stopped interviewing participants once we reached thematic saturation, a common practice in qualitative analysis [65].

**Qualitative Analysis.** We used a combination of deductive and inductive qualitative analysis to understand the experiences of BLV users with account creation forms and authentication (RQ3). The first author developed an initial deductive codebook from the web

**Table 1: Mapping Quantitative Measurement Issues and Qualitative Findings**

| Issue                                                | WCAG Codes                                                                                | Description                                                     | Qualitative Experiences                                                                          | Our Novelty                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Issue 1: Inaccessible Labels</b>                  | 4.1.2, 2.4.4, 3.3.2, 2.1.1, 2.4.7, 2.5.3, 2.4.3, 3.3.4, 3.3.8, 3.2.4, 1.3.5               | Unlabeled fields/links; unclear instructions.                   | P(1, 2, 4, 6-10, 12) struggled with field info, leading to trial-and-error/frustration.          | [3] only looks at 37 auth. schemes, [4] briefly discusses this and does not measure it, [87] looks at labels in the context of ads, and [23] is over a decade old, and only qualitatively measures these concerns.                                                                                                                                  |
| <b>Issue 2: Min. Feedback &amp; Inaccess. Errors</b> | 4.1.2, 4.1.3, 1.3.1, 1.3.3, 3.3.3, 2.1.1, 2.4.7, 3.3.1, 4.1.2, 2.4.3, 3.3.8, 3.2.6, 1.3.5 | Error/status updates not announced to screen readers.           | P(1, 5, 8, 11) faced generic/unspecific errors, forcing page re-reads or guessing.               | [23] is over a decade old, and only qualitatively measures these concerns, and [61] only briefly mentions this.                                                                                                                                                                                                                                     |
| <b>Issue 3: Inaccess. Buttons/Checkboxes</b>         | 4.1.2, 2.1.1, 4.1.2, 2.4.3, 3.3.8, 1.3.5                                                  | Keyboard-inaccessible or incorrectly announced buttons/toggles. | Inaccessible buttons P(4,7-9,12) and checkboxes P(4, 6, 7, 9, 10, 12) caused data/privacy risks. | [87] looks at buttons in the context of ads, [23] is over a decade old, and only qualitatively measures issues with buttons, [18] measures color contrast and alt text in the context of cookie notices, [24] tersely mentions this in the context of general UI design, and a participant in [53] briefly discusses inaccessible button placement. |
| <b>Issue 4: CAPTCHAs</b>                             | 1.1.1, 3.3.8                                                                              | CAPTCHAs lacked usable or clear audio components.               | P(2, 4, 5-9, 11-13) were blocked from account creation/authentication.                           | CAPTCHAs were highlighted in [17, 24] to be a significant issue, and [33] is a broad overview of CAPTCHA inaccessibility by the WCAG.                                                                                                                                                                                                               |
| <b>Issue 5: Inaccessible Password Meters</b>         | 4.1.2, 2.1.1, 2.4.7, 2.4.3, 3.3.8                                                         | Password strength meters were only visual.                      | P(1, 2, 7-9, 11) ignored meters, hindering informed password composition.                        | Meters were briefly mentioned in [24] as both limiting and sometimes helpful.                                                                                                                                                                                                                                                                       |
| <b>Issue 6: Password Policies</b>                    | 4.1.2, 1.3.1, 2.1.1, 2.4.7, 2.4.3, 3.3.8                                                  | Requirements shown only after reload/error.                     | P(1-3, 5-8, 10) frustrated by negative feedback loops and lost inputs.                           | –                                                                                                                                                                                                                                                                                                                                                   |
| <b>Issue 7: Poor Design Choices</b>                  | 1.1.1, 4.1.2, 3.2.2, 2.1.1, 2.4.7, 1.3.2, 2.4.3, 2.1.2, 3.3.8                             | Pop-ups/promos divert focus or cause unintentional sign-ups.    | P(4, 7, 8, 10-12) noted focus interference and privacy concerns.                                 | [23] is over a decade old, and only qualitatively measures issues with pop-ups, not other promotional fields.                                                                                                                                                                                                                                       |

measurement results, which was further refined by the second author’s input [14, 75]. Next, the first author open-coded the first two interviews using MaxQDA [76] to generate a preliminary set of codes. These inductive codes were then combined with the deductive codebook to form an integrated codebook. The first and second authors double-coded three interviews using the integrated codebook. After each iteration, the authors came together to discuss new additions and resolve conflicts, either by removing redundant codes or adding new ones [75]. After they reached an inter-rater reliability (IRR) of  $\kappa = .84$ , which indicates high agreement [51], the two authors split the remaining transcripts to code separately.

**Recruitment and Demographics.** We recruited participants through the National Federation of the Blind (NFB) mailing list, and compensated participants for their time with a \$25 USD Amazon gift card. The Open Science repository 8.2 provides demographic details of the participants. Our sample consisted seven women, five men, one agender individual, and one non-binary participant. The interviews took place between September and October 2025.

## 4 LIMITATIONS

We begin by outlining the limitations of the paper’s quantitative section, specifically noting that we only measured a small subset of the global web ecosystem. This is compounded by the fact that some parts of our analysis were unable to capture the full range of 20,120 websites from Alroomi and Li [6], either because they were not originally authentication forms or because our scraper could not access the website. We randomly selected 100 websites and subsequently performed a manual audit of 325 pages to check for false positives of non-login or sign-up pages. While our manual audit ensured there were no false positives in our dataset, some may still remain due to the large volume of websites collected. Relatedly, we only performed a high-level analysis of inaccessibility patterns across website categories as part of RQ1, RQ2. Furthermore, we

chose to manually audit only a small fraction of login and sign-up forms, with only one author audited both interfaces. This was to prevent a less-trained evaluator from missing issues, since only one member of the research team had professional experience in accessibility auditing. We aimed to make the manual audits as representative as possible by auditing both popular and less-popular websites equally. We also note that our interviews only looked on a subset of disabilities (BLV users) when the WCAG-informed quantitative analysis focuses on disabled users more broadly, which limits the interpretability of our qualitative results.

Additionally, our qualitative analysis has some limitations. First, while we did aim to recruit participants with diverse age and educational backgrounds, the majority of our participants were blind from birth, so we cannot say that our results are generalizable to the full spectrum of vision differences. Likewise, our results are US-centric, and likely do not capture the complexities of web accessibility globally. Finally, our work only captures a small subset of potential issues related to authentication; we decided against going into detail about things such as account management, sharing, and other security concerns to keep the interviews as focused as possible. We urge researchers to explore these issues more thoroughly.

## 5 RESULTS

This section presents the findings from both the quantitative measurements and qualitative interviews. To better illustrate how the quantitative (breadth) data connects to the qualitative experiences of BLV users (depth), we group categories of common issues that are present in both the web measurement and expressed during interviews under 7 issues. Category names and descriptions come from our data directly, and Table 1 describes these issues and their presence in prior work.

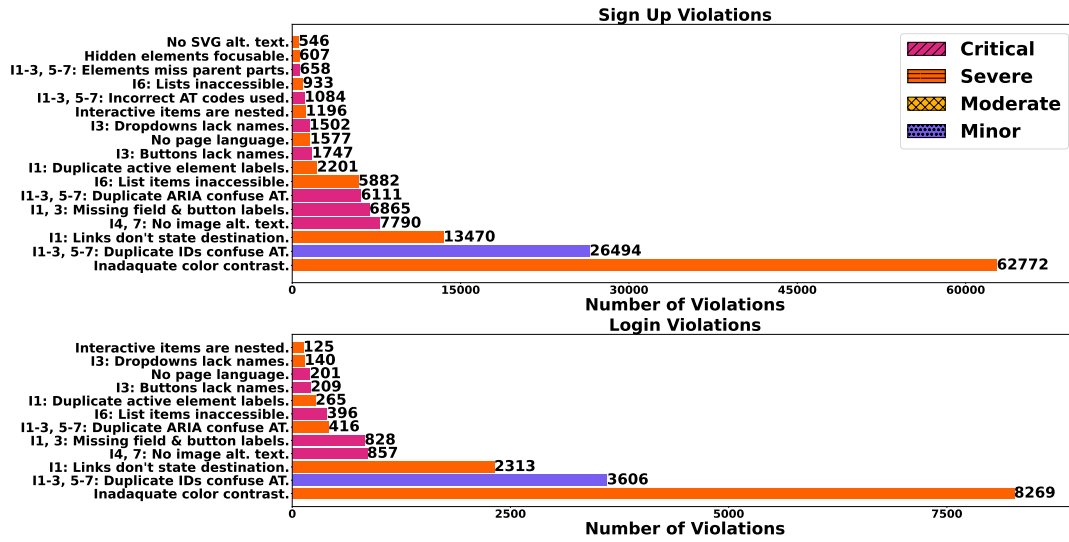


Figure 1: Bar graphs numbering the amount of sign-up and login violations. The y-axis includes the issue's associated mapping in Table 1. For visual clarity, violations with less than 100 instances are omitted from the graph.

## 5.1 Quantitative Website Accessibility Audit

This section presents our quantitative results during both the manual and automated audits (RQ1). We also go over key issues identified of broad concern during our audits (RQ2). Throughout this section, we map the specific issues identified in our audits to Table 1, which connects these findings to the interview results discussed in our later qualitative section and to prior work. In total, there were 159,060 individual errors automatically detected (Figure 1). 81% (128,960 out of 159,060) of the errors are labeled as **critical or serious**, meaning that they pose significant barriers to people with disabilities. Axe-core defines both critical and serious errors as ones that make it near-to-totally impossible for users to access critical information [42]. On account creation forms, this increases the likelihood for users to abandon workflows or make decisions that put their security and privacy at risk [19, 31] due to the sheer amount of errors that users will have to negotiate with [28]. Likewise, we also saw a total of 2,866 individual issues identified on 325 websites during the manual audits.

**Inaccessible and Unobservable Elements.** Inaccessible elements present significant security harms. For example, **Issue 1: Inaccessible Labels** in Table 1 (labeled “Provide form labels” in Figure 2), with 576 total instances for manual audits, makes it difficult for users to ascertain what information is required to enter into form fields, adding an additional cognitive burden to assistive technology users. This is also reflected in our large-scale audits with the “Missing field & button labels” in Figure 1, with a total of 7,693 errors. Further, these errors may lead users to enter sensitive data, like passwords, into insecure plain-text fields; this is particularly realistic given that our audit targeted only sign-up and login forms.

Another example is the link-name error (“Links don’t clearly state destination”, Figures 1, 2), which occurs when a link lacks a clear purpose, is inaccessible to screen readers, or cannot be focused via keyboard navigation. Without a clear destination context, users

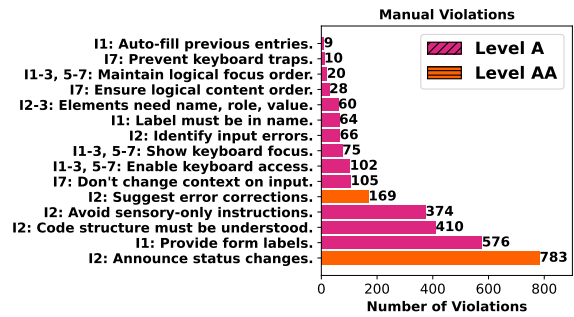
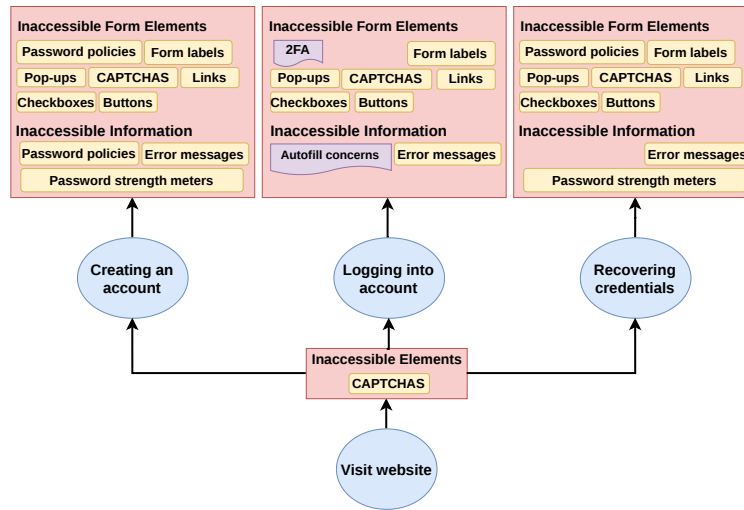


Figure 2: Bar graphs numbering the amount of sign-up and login violations identified during the manual audit. The y-axis includes the issue's associated mapping in Table 1. For visual clarity, violations with less than 8 instances are omitted from the graph.

who rely on screen readers cannot easily predict where a link leads, which can result in a clickjacking attack, where users are navigated to unexpected or unsafe pages.

We also identified over 400 instances (labeled “Code structure must be understood” in Figure 2) of account creation pages that contained both an email form field during sign-up and a promotional “sign-up for our newsletter” email input field at the bottom of the page during our manual audits, which leads to ARIA overlap. When two elements have the same ARIA identifier, it will be unclear to the assistive technology (and user) that the elements are different – it could cause screen reader users to unknowingly share their email with the newsletter team. This poses a direct privacy risk, since the website is using their data without explicit consent. We also label these errors as **Issue 1: Inaccessible Labels** in Table 1.

We also saw a substantial number of inaccessible buttons and checkboxes on sign-up and authentication forms (**Issue 3: Inaccessible Buttons/Checkboxes** in Table 1), including password



**Figure 3:** A diagram of accessibility concerns that BLV users can experience when creating and authenticating into a website. The purple issues, outlined in a rectangle with a wavy bottom, are from the qualitative section, while the other issues are supported by both the qualitative and quantitative sections.

visibility buttons and promotional sign-up checkboxes. Some forms simply did not allow keyboard selection (“Enable keyboard access”, Figure 2 for buttons or checkboxes; while this undoubtedly impacts accessibility, it also has security and privacy implications beyond inconvenience. As prior work has highlighted, a user may decide to use a weaker, but more easily-typed password [34, 74] in place of a stronger one if they cannot toggle on password visibility (and cannot view what they are typing). Similarly, unlabeled consent checkboxes expose screen reader users to privacy risks, such as inadvertently opting into promotional emails due to a lack of accessible context – this particular scenario is explicitly raised during our interviews. More obvious accessibility issues were also present on account creation forms, such as **Issue 4: CAPTCHAs** (Table 1). For example, we observed that some websites required the auditor to complete an embedded, inaccessible CAPTCHA to continue with the sign-up process. These CAPTCHAs did not provide any audio feedback, leaving blind users without a way to complete the account creation process. This follows broader CAPTCHA trends being widely inaccessible on the web [33]; many of our participants discussed inaccessible CAPTCHAs making it difficult to complete authentication-related tasks.

**Inaccessible, Unobservable Feedback & Errors.** We found that authentication forms did not always provide users with feedback in an accessible or convenient way (**Issue 2: Minimal Feedback & Inaccessible Errors**), this error relates to the WCAG criteria 4.1.3 Status Messages. This issue accounted for 783 individual instances (labeled “Announce status changes” in Figure 2). This criterion requires that changes in content, such as error messages, are programmatically determinable – that is, assistive technologies can read and announce them. To fully comprehend this issue, it is important to note that screen readers operate in two modes: browser and form mode [72]. In browser mode, the entire page – including

paragraph text, headings, and other non-focusable elements – is read to the user. In form mode, the screen reader only reads focusable text, such as labeled paragraph text, links, form fields/elements, and buttons. These modes do not operate in unison and require the user to switch between them.

While most error messages and password requirements were available to the auditor at some point during the manual audit (either before or after entering dummy information), they were only available in browser mode, and not embedded as interactive elements. This requires screen reader users to switch between form and browser mode to interact with authentication form elements and their requirements. While this issue does not directly compromise security choices, it does increase cognitive burden and risks information fatigue for BLV users [28]. In the worst case, for example, this means that there would be no indication to screen reader users whether the content they submitted was valid, unless they re-read the entire webpage – a user could submit a password that is non-complaint, and only find out about that error once they re-read the page again.

**Password Policies & Meters.** During our manual audits, we observed that the majority of PWP were delayed, meaning that screen readers users would have to edit their password only after the account creation page had reloaded. Furthermore, many of these notifications were only visible in browser mode, not form mode. This means that a user must enter erroneous information before understanding the requirements of a given form, while simultaneously switching between browsing and form modes. This creates a negative feedback loop where frustration, inaccessibility, and security fatigue can result in users giving up and choosing the most convenient option (such as reusing an old password) over the most secure one [19, 37, 41]. We label this issue as **Issue 6: Password Policies** in Table 1.

Password meters were also rarely accessible via screen readers and only presented a visual indicator of password strength. This, again, prevents screen reader users from making informed decisions about their password composition [73], which directly impacts the security of an account. Ideally, strength meters should have an **unintrusive** visual component, text component, and sound-based component, similar to how progress bars are indicated to screen readers [52]. This is summarized as **Issue 5: Inaccessible Password Meters** in Table 1.

**Poor Design Choices.** We now highlight various design choices that could influence a user’s security decisions. First, we found that many websites displayed pop-ups during authentication; beyond being inconvenient, these pop-ups embed extra inputs (such as email fields) for promotional sign-ups (**Issue 7: Poor Design Choices**). While not directly impacting security behavior, these interruptions present a significant challenge for screen-reader users, who may become disoriented when their focus is diverted while interacting with an account creation form. Even for pop-ups with appropriate alert tags, a user may not notice this warning if they are already actively navigating the form fields within the pop-up itself; that is, a user may skip the alert because they are already in the process of tabbing through the form. Relatedly, we also saw that websites had unlabeled promotional fields within their footers, which could also present issues to screen reader users. Both design patterns heighten the risk of users unintentionally entering personal data into a promotional field, impacting the privacy of the user. While contextual indicators such as icons and descriptive text may allow users to discern where they are on a page, websites should not assume universal comprehension with just these clues [53].

**Website Classification.** From our website categorization, we find that across all top categories, inadequate color contrast remains a dominant violation; this is unsurprising, since websites tend to include many kinds of visual elements. Axe-core classifies these issues as severe. While prior work highlights how inaccessible elements, including color contrast issues, impact fatigue [28], low contrast can obscure parts of websites, including security and privacy related information such as account warnings, form instructions, and link destinations. Likewise, we also see a large amount of missing image alternative text and duplicate IDs, supporting our findings in both our quantitative and qualitative analysis that users relying on assistive technologies face structural hurdles that likely disrupt their ability to create or access accounts. On average, each website category had at least 2 unique violations per website in every category (Appendix 8.2), indicating that these websites likely have a cluster of accessibility issues that make safe interaction with sign-up and login forms extremely strenuous for these users.

**Summary of Quantitative Results.** We highlight seven patterns of inaccessibility during account creation and authentication for BLV users (see Table 1). These include incorrect labels or instructions, minimal feedback and errors, delayed PWPs, inaccessible buttons/checkboxes, inaccessible CAPTCHAs, inaccessible password meters, and poor design choices, such as pop-ups. The prevalence of these issues on the web is high, with over half of the issues measured making it difficult for users to access information

during account creation and authentication. While the quantitative measurement establishes the frequency of these inaccessibility patterns, it does not capture how these issues hinder the security and privacy choices of BLV users and what strategies they employ when navigating these issues. As such, we conducted qualitative interviews to highlight how these inaccessibility issues impact users’ experiences with account creation and authentication.

## 5.2 Interview Study

Building on the insights from our quantitative work, we find that the majority of our participants experienced accessibility issues related to navigation, information retrieval, and information verification during account creation and authentication. Interviews also highlighted barriers that the web measurement alone could not have elucidated, such as difficulties using 2FA [4], recovering their accounts, and using autofill with their assistive technology. This underlines the tensions between inaccessible account interfaces, and users’ lived experiences with the insurmountable security and privacy risks they generate. Furthermore, we also explore the strategies and tools our participants use during account creation and authentication for dealing with inaccessible elements.

We break down this section into four subsections, corresponding to our broader code themes. Notably, we divide this section into two subsections: the first part details the issues that directly map to the quantitative results in Table 1, the second for those that our quantitative measurement did not capture. We then highlight the strategies that our participants use to navigate inaccessible account creation forms and authentication, and their perceptions of these tools; for example, some of our participants use and appreciate the convenience of security tools such as password managers [61], passkeys, and ad blockers [87]. Other participants, in contrast, have negative accessibility experiences with these tools, such as having to go back to verify proper autofill, suggesting a gap in current implementation. We conclude by presenting participants’ privacy concerns with the tools that they use during account creation and authentication.

### 5.2.1 Issues With Form Elements, Feedback, and Navigation.

**Issues 1, 2, and 3: Inaccessible Elements and Errors.** In the first section of the interview (III), we ask participants about their broader credential-creation and inaccessibility experience. We observed that a number of our participants experienced issues with form field inaccessibility, particularly with links and form fields (**Issue 1: Inaccessible Labels**), error detection (**Issue 2: Minimal Feedback & Inaccessible Errors**), and buttons (**Issue 3: Inaccessible Buttons/Checkboxes**). For example, P1 describes having trouble discerning what information to enter into forms and how to interpret the resulting error messages (**Issue 2: Minimal Feedback & Inaccessible Errors**):

*“And a lot of times, the form input fields were not properly labeled, so it just took me a long time to fill in the correct information. I’ve seen websites where I’m trying to create an account, and it would just say error, but it wouldn’t say what kind of error it was, or which field it was for.” – P1*

Furthermore, we also see that our participants express annoyances with other inaccessible elements. For example, P4 discusses a pattern they observe with inaccessible buttons on account creation forms:

*“The other problem that I run into is the ever-evasive button. All it says, button. Next one. Button. Button. Button. They’re unlabeled, and it turns into this guessing game. I know one of them is submit. I have to click on all of them. One of them deleted all my responses. One of them brought me to a completely different page ... but you’re doing all these other things that you don’t know what you’re doing, and it’s because the only remedy is to figure it out.” – P4*

Because of button inaccessibility, P4 must use trial-and-error to figure out how to navigate an account creation form, putting himself at risk of a clickjacking attack. Links presented similar issues to our participants as well (P7, 8, 12). A similar sentiment is shared by P7, who comments on unlabeled checkboxes: *“And sometimes you don’t know if you’ve checked something, and then you can find out later. ... that can really have bad consequences. If something’s checked that you want it unchecked”*. This highlights the potential privacy violations that inaccessible checkboxes can perpetuate; indeed, many of our participants stated that they have similar issues with checkboxes (P4, 6, 7, 9, 10, 12).

**Issue 4: Inaccessible CAPTCHAs.** Many of our participants (P2, P4-P9, P11-P13) raised concerns with CAPTCHAs. As P13 explains, *“[CAPTCHAs], that does bother me, and when I have to check the boxes because I can’t see them”*. Even when CAPTCHAs have an audio component designed for BLV users, participants have found that they are sometimes too muffled to understand, or do not work at all.

*“Oh, they’re so bad. Well, and they have audio challenges, but half the time they work and half the time they don’t. And if you can’t get through the CAPTCHA, you’re, like, you’re stuck. Like, pick all the images with street lights. Mmm, no.” – P4*

In summary, the implementation of these security mechanisms leads to a persistent pattern of inaccessibility when disabled users encounter them, which prior work has catalogued extensively [13, 23, 24, 33, 53]. Even when CAPTCHAs are designed with disabled users in mind [13], our work and prior research has highlighted how these systems still continue to fail these users, which can result in an asymmetric level of burden compared to non-disabled peers.

**Issues 5 & 6: Password Strength Meters & Policies.** Participants were also asked about password policies (PWPs) (IV) and password strength meters (V). In particular, password strength meters often posed difficulties for some of our participants, leading them to ignore them entirely when creating passwords (P2, P7). P11 expresses annoyance with how password strength meters’ decelerations of strength during password entry interrupt her, expressing how they cause her fatigue:

*“Yes, strong, medium, yeah, while you’re typing, before you even finish. And then after you’re finished. Because it’s talking as you are typing, after you’re finished, you*

*forgot what it says. And then you have to, like, navigate and have it repeat.” – P11*

Participants also voiced considerable frustration with many aspects of the PWPs. Participants (P1, P2, P13) who use personal word systems to generate passwords expressed significant grievances with having to account for the policies of websites – this is in line with previous findings, which has highlighted how policies have noteworthy implications for the security of users’ accounts [41, 66]. Participants (P2, P8, P11, P12) also expressed issues with delayed PWPs.

*“Yeah, so I will fill it out. And I know [the password field] would say, ‘error’. And then I would have to navigate to the place where it tells you the password requirements, like, it ... doesn’t tell you at the beginning. Only after you enter, and it’s incorrect. So then I’ll have to ... go to that section. And then think of a new password, and then go back to entering the password. And if we refresh the page, it’s gone.” – P11*

Along with P11, P2 also expressed frustration with how page refreshes remove the users’ previously-entered content, indicating how these systems fail to both provide persistent feedback **and** trap users in a cycle of trial, error, and re-entry.

**Issue 7: Poor Design Choices.** Participants also expressed concerns with other elements, such as pop-ups (P4, P7, P8, P10-12), in line with both our measurement results and prior work [23]. When asked broadly about their biggest authentication-related problem they experience on the web, P8 explicitly notes pop-ups:

*“I would think the thing that’s probably most annoying to me is the pop-ups ... That’s a big thing, and sometimes, they have all these graphics up there, so your screen reader can’t read all the things it’s supposed to be able to read. And that can be very frustrating.” – P8*

In both physical [15] and digital spaces, BLV users must strain against inaccessible systems to ensure their safety [23, 53], a friction that, aligning with our quantitative findings, heightens information fatigue and choice paralysis during authentication.

### 5.2.2 Other Accessibility Concerns.

**Password Reset & Recovery.** In the face of forgetting a password, users may elect to instead reset their password. Users generally have to use a recovery interface to do so, where the website provider requires the user to reverify their identity – this is sometimes done by sending an email to the user’s account, or sending a verification code to another device. Because password reset and recovery often involves interacting with authentication interfaces, we were curious to understand users’ experiences with these methods more broadly (Section IX in the Interview script). P2 and P11 both expressed frustration with resetting their passwords, particularly on websites they use infrequently or on those with stringent verification requirements, such as banks. Inflexible authentication mechanisms have been shown to significantly affect disabled users’ access to essential services [8]. Recovery mechanisms posed similar barriers, introducing security risks alongside usability hurdles when repeated access failures triggered account lockouts:

*“It would say something like ‘type in the last password you remember’, but that didn’t work. So then I would click on, like, ‘try another way’ a bunch of times, and then it got to a point where they were, like, you’ve tried all the ways, and you will need one of our team members to sign you in. It says something like that, and I have to wait a week before someone can help.” – P1*

As highlighted across our findings, disabled users are often forced to recover their credentials simply because their authentication mechanisms or reset functionality is inaccessible to them [23, 24].

**2FA.** Given prior research showing 2FA challenges for blind/LV users, we surveyed participants’ 2FA experiences (Section VIII), finding that 2FA presented challenges to some of our participants (P1, P3–P5, P7, P8, P11, P12). When discussing verification code entry, P8 expresses:

*“Sometimes [screen reader] doesn’t see the codes, or sometimes I might know where the code is, but I don’t know where to put it. Because it’ll be all in one field. Like, you have 6 digits or 7 digits, and they’re all in one field. But some, you have to do each digit in a separate field.” – P8*

Furthermore, P7 describes having been forced to use Microsoft Authenticator’s QR code scanner as a second factor for her work account, which she was unable to do because she could not visually perceive the QR code. Due to these setup and interaction difficulties, multiple participants reported generally skipping 2FA setup, even when it was recommended. This leaves their accounts exposed to single-factor vulnerabilities.

We note that not all of our participants reported the same level of inaccessibility with both account creation and authentication; our participants’ experiences were varied and non-monolithic. For example, P13, who has limited vision, reported having no issues with inaccessible elements. P5 mentions that they usually only visit popular websites, so they do not encounter as many issues compared to other participants. Instead, we aim to showcase how various, compounding elements can significantly hinder users’ decision-making processes; not only can inaccessibility disincentivize usage of security tools (such as 2FA) they can also be actively dangerous (unlabeled buttons, links). In the following section, we describe our participants’ strategies for dealing with inaccessible elements of account creation and authentication.

### 5.2.3 Strategies to Mitigate Inaccessibility Features.

**Forms.** Along with asking about the specific accessibility violations, we also inquired about participants’ strategies for overcoming these obstacles (Section III). When dealing with inaccessible account creation forms, participants expressed that they would often have to resort to trial-and-error (P7, 11). As outlined in the previous section regarding buttons and links, this can result in security or privacy compromise. Participants (P3) also use website context clues to help them fill out account creation forms; for example, even when a form field is not labeled as a password field, if the screen reader censors the entered text, users assume that the field is for passwords.

Participants also use close allies (family and friends) and services like Aira [2] and BeMyEyes [25] for account creation assistance. P8, for example, comments on his experience using Aira to help

identify error messages: *“I’ll go through and go to Aira, and they’ll tell me. But it’s like, I shouldn’t have to do that”*. As highlighted by prior work [5] and our interviews, however, our participants have privacy concerns with giving strangers personal information about their surroundings.

**Autofill.** We further ask participants about their experiences with autofill (VII), with all of our participants reported using their browser’s autofill or password managers to aid them in form entry, finding them both helpful and assistive. In conjunction with autofill, one participant (P4) also remarked that their screen readers often auto-focus on the autofilled information, making it easier for them to know that the filled information is correct. Notably however, some participants also remarked that they often need to go back through the form to ensure the correct information has been filled in the correct places, which is a known issue with password managers and autofill [35, 36]. When not relying on autofill, one participant (P1) noted that they choose to simply reset their passwords.

**Password Masking & Policies.** While prior work has found that users are generally accepting of password masking [34], where input characters are replaced by symbols (e.g., asterisks), we found that masking often presented many difficulties to our participants (P1–P3, P5–P8, P11–P13). As outlined in Section 5.1, many of the websites we analyzed did not allow keyboard selection of password masking toggles. As explained by P2, their assistive technology is often incompatible with password entry fields when password masking is not available:

*“Like, the braille keyboard, it can be very finicky sometimes. If you can’t hear what you’re typing, you just don’t know that it’s doing it. Passwords are very hit or miss with braille screen input.” – P2*

Due to the positioning of braille characters relative to keyboard characters, it can be very difficult to know if a password is being typed correctly when password masking is not optional. However, we also note that participants (P8, P11) also choose to use their braille keyboards instead of the standard keyboards as both privacy preservers (discussed in the following section), and because they find them easier to use when faced with masked passwords. Like sighted users [74], participants will also choose to edit their passwords in-place to meet the needs of PWP – this can be both frustrating and inaccessible, as outlined in Section 5.2.1. To further accommodate inaccessible password masking and policies, some of our participants elect to write their passwords down in plain-text documents, such as an Excel spreadsheet or their Notes app, and then copy-paste them into form fields. This leaves them vulnerable to credential theft and account compromise [7, 47].

### 5.2.4 Security and Privacy Aids and Concerns.

**Aids.** Participants employ various aids to enhance their security and privacy, including Apple’s randomized email feature (P7) [9] and password managers (P2, P4, P5, P11). Many view their assistive technology suite as tools that preserve their security and privacy, but also consider their security and privacy tools as assistive technology. For example, P2 explains how they consider their ad-blocker essential for their web-browsing:

*"I use a lot of ad blockers as an accessibility feature. People don't understand how much an ad can absolutely destroy your ability to use a website with a screen reader. Because it will just latch onto your screen reader and won't let go, and you can't do anything, and you can't close it." – P2*

Interestingly, this is in contrast with the findings of Yeung et al. [87], who found that participants often forgo using ad blockers due to usability issues. Both P2 and P4 highlight how their phone's screen curtain prevents sighted people from seeing their activity, and for P4, the high-speed output of his screen reader makes the audio sound incomprehensible to others. These mechanisms demonstrate novel ways disabled users safeguard their privacy.

**Concerns.** While many of our participants use their browser's autofill or password manager to help with account creation and authentication, some participants (P3, P4, P8, P11) also have reservations about using these services germane to security and privacy implications, especially regarding breaches. Indeed, password managers have been breached in the past [64]. However, they also concede that the convenience outweighs the potential risks: *"I don't know how much more unsafe it is by using autofill, and I have some thoughts about security, but in the end, this is more convenient"* (P11). Crucially, in regards to non-optional password masking, some participants (P7, P8) forgo creating strong passwords entirely so they can adequately remember the passwords they set for their account. Some of our participants also report mistrust when relying on allies. For example, when asked whether P13 ever reached out to a bank to help her reset her password, she remarks: *"No. And I wouldn't trust them, because people are people"*. Services such as BeMyEyes and Aira were also criticized. P8 expresses: *"It's fine for a game site or something like that, but if you're doing something else, you probably should not use them"*. This sentiment is also shared in prior work [5].

## 6 DISCUSSION & CONCLUSION

We conducted a mixed-method study to identify accessibility issues in account creation and authentication processes, including an audit of over 11k sign-up and login interfaces. We identified seven recurring accessibility issues, specifically **Issue 1: Inaccessible Labels**, **Issue 3: Inaccessible Buttons/Checkboxes**, **Issue 4: CAPTCHAs**, and **Issue 5: Inaccessible Password Meters**. Together, these findings highlight that inaccessibility issues are highly pervasive and can make account creation and the login process extremely tedious for the BLV users. To understand user struggles holistically, we interviewed BLV users to see how audit-identified issues impact their daily sign-up and login experiences. We found that our participants experience disproportionate mental load and friction [28], with every participant having experienced some accessibility issues identified in the website audits at some point. Beyond confirming the quantitative accessibility issues, our participants also revealed more nuanced challenges that standard audits missed, such as password recovery, error formatting, and 2FA. Figure 3 illustrates where these issues are likely to appear during account creation and authentication.

Some participants also reported positive login experiences when using browser's and password manager's autofill feature, as it made

it easier to enter correct information while working seamlessly with their screen reader. The security benefits of other tools, such as ad blockers, recovery mechanisms, and masking were also expressed during our interviews. Participants also used novel strategies to navigate the accessibility hurdles identified in the interviews. While many resorted to trial and error for form-related issues, others relied on third-party services [5] such as Aira [2] and BeMyEyes [25]. Participants also leveraged assistive technologies (AT) and external allies to manage authentication, underlining the vital role of these resources when interacting with disabling environments [11].

### 6.1 Shifting Trends in Authentication

Authentication, and usable S&P as a whole, has changed substantially since Dosono et al.'s paper in 2015 [23]. Since then, the rise in biometric authentication [45], passkeys [43], and MFA [50, 57, 63] on websites has given users more options on **how** they want to authenticate, with even password guidance [54] having adapted to account for better usability. Privacy control mechanisms, such as dashboards [26] and labels [39, 86], have also seen increasing adoption within this ecosystem. However, despite over a decade of research on the inaccessibility of S&P mechanisms [3, 4, 8, 15, 18, 23, 33, 86, 87], our results indicate that even benign systems still hinder BLV's capacity to meaningfully engage in S&P decisions, notwithstanding the accessibility-focused sentiment present within our community for over a decade [23, 80]. More work is necessary in understanding how to design authentication, and S&P mechanisms as a whole, to better meet the needs of BLV users. Particularly for the disabled community, authentication mechanisms have posed monumental barriers, primarily due to AT incompatibilities [8, 45] **and/or** underlying assumptions regarding user cognitive capacity [58]. New approaches to authentication should consider broader human experiences, whether they are mediated through AT or otherwise, that are outside normative body-minds [11, 80]. Likewise, from our results, we hypothesize that other, more complex S&P mechanisms (i.e. labels, dashboards, etc.) [1, 18, 26, 39, 86] are structurally inaccessible to disabled users. This requires further investigation to better understand how to best support at-risk users [69].

### 6.2 (In)accessibility Impacts Security & Privacy

Our results signify a portend of growing inaccessible S&P systems, and speaks to the broader systemic issues regarding access, where disabled users must accommodate systems inherently not designed for them [11] or their AT. As a result, they are forced to take actions they would not otherwise take that can put their S&P directly at risk, or experience increased fatigue, which impacts their S&P decisions. Likewise, disabled users have minimal flexibility to adjust these systems compared to able users. Sighted users could ignore password strength meters [73] or find no issue acknowledging the password policies [41] when they appear on the account creation interface. However, both of these elements often interrupt the password entry process for BLV users due to how screen readers interact with webpages. This forces them to re-read the entire page from the beginning, making the process extremely strenuous, cumbersome, and adding to information fatigue due to how the

security information is presented, and then processed, by screen readers [28].

A key distinction between our work and prior research is the systematic characterization of accessibility issues that precipitate S&P concerns. We show that these are not only access-related challenges, but also sources of harm that the S&P community must actively address to mitigate risk. When form fields are not labeled properly, users may enter sensitive information into plaintext fields or into fields they had no intention of interacting with. Or, when pop-ups arrest users' keyboard navigation, they might not realize they are entering information into a pop-up. When form elements are not keyboard-focusable, users who rely on keyboard navigation, such as users with mobility disabilities [45], are effectively locked out of performing actions that able-bodied participants can accomplish. Without accessible password policy information, users with information processing disabilities [58] will likely struggle with creating secure passwords or solving CAPTCHAs. Furthermore, mediating interactions through proxy systems (AT) or third parties can inadvertently trigger secondary harms. AT with longer input speeds, such as sip-and-puffs and eye tracking systems, can lock users out of time-based authentication mechanisms, at no fault of their own. As prior work has outlined, text-to-speech, speech-to-text, and other assistive services [5, 8] can inadvertently expose users' information to bystanders. When all of these issues compound, it can make the process of account creation and authentication a serious S&P risk and lead to decision fatigue, where the quality of a decision deteriorates with the more information a user has to process; for systems that rely on critical choices (such as S&P systems), this can have disastrous consequences, such as data exposure, unauthorized account access, and complete abandonment of the service. In turn, this disincentivizes users from engaging with security tools, often at no fault of their own. A similar pattern exists with other S&P tools, such as ad-blockers [87], email phishing warnings [88], cookie notices [18].

Preventing decision fatigue is one of the key motivations for the adoption of privacy labels [39, 86] and nudges in S&P systems [1] – if we are serious about mitigating these harms, we must address how persistent inaccessibility systematically manufactures these risks for disabled users [23, 45, 87]. We emphasize that many of the issues touched upon are complex, and likely require wholesale paradigm shifts toward more inclusive social cybersecurity awareness [12, 62, 85].

### 6.3 Improvements to Authentication

A concerted effort, across multiple aisles, is necessary to make meaningful changes in authentication experiences. To address these concerns, we present specific stakeholder recommendations. Importantly, underlying all of these recommendations is the necessity of involving diverse disabled perspectives throughout the development process, echoing the motto of “nothing about us, without us”<sup>2</sup>. Outside of direct involvement [89], this can be done through co-design, working groups, or advisory boards. As Zong [89] and Hofmann et al. [32] contend however, indirect participation often

does not address the systemic issues (ableism) of a system, but instead only addresses a symptom of that system (“impairment” [32], or inaccessibility). Echoing critical disability/HCI researchers, disabled knowledge production must be at the forefront of developing equitable S&P paradigms; pragmatically, this can take the form of sponsoring disability-first initiatives, co-organizing with disabled people, allying with disability justice scholars [32], or simply letting disabled people take the lead on projects.

**User experience (UX) designers.** In this section we make recommendations to UX developers and designers. Critically, accessibility considerations must be at the forefront. Instead of bootstrapping accessibility after development [12, 62], designers should consider disabled perspectives before, during, and after project development. Hiring disabled quality assurance testers can identify barriers that automated tools [67] or non-disabled experts cannot detect. For UX designers, along with making accessibility a priority, we make the following recommendations, which follow directly from our results:

- **Password policies:** We recommend that PWPs are displayed to users in advance of password creation, allowing them to internalize the requirements, as well as utilizing contextual error messages.
- **2FA:** Users should be offered several 2FA [3, 45] choices that give them the flexibility to pick the option they find most suitable to their needs.
- **Pop-ups:** To prevent fatigue and distraction, account creation pages should be restricted to essential authentication elements, excluding all non-relevant promotional content or pop-ups.

Lastly, we found that many of the issues we observed during our audits can be resolved by following WCAG AA guidelines [79], such as proper form labeling and keyboard navigability. We recommend that developers find ways to integrate AA compliance directly into workflows, such as accessibility flags that warn developers when non-compliance occurs before it reaches production.

**Security & Accessibility Practitioners.** Security and accessibility guidelines are often in contention, such as NISTs' guidance [54] on 1 hour session timeouts directly contradicting WCAG's guidelines on 20 hour timeouts [78]. Even outdated and inaccessible NIST advice, such as consistent password changes, still permeate in on-line discussion [40], and recommended by experts [30]. To address these discrepancies, we recommend direct collaboration between accessibility and security practitioners to establish shared compliance standards, similar to the WCAG [79]. This can include collaboration with community groups, workshops, or digital consortia. This, in turn, can support the development of accessible authentication interfaces, standards, and paradigms. Furthermore, our findings suggest that participants see their AT as privacy preservers, yet often come into conflict with account creation interfaces. Consequently, practitioners should explore how to optimize AT design to aid in authentication. S&P practitioners should also consider disabled users' experiences with S&P software, such as ad-blockers [87] and email phishing warnings [88], when deciding what safeguards should be implemented. Finally, accessibility regulations are not effective if they are not enforced. We urge policy-makers and regulators to

<sup>2</sup>This is a phrase from the disability justice movement that emphasizes the need to involve disabled people in projects and decisions that involve them.

push for stronger accountability measures for digital accessibility violations, such as specialized digital accessibility bureaus, so that enforcement does not rely on individual court cases to be remediated [29]. This will not only improve accessibility, but also S&P outcomes.

**Researchers & Academics.** Researchers can assist in designing disability-friendly account creation elements, such as password meters and error messages, particularly for screen reader users. In particular, we highlight the frustration users still have with CAPTCHA systems [33]; with generative AI's growing aptitude at solving them [60], we recommend that researchers further explore potential avenues of alternative and accessible assurance mechanisms. One way this can be done is by evaluating mechanisms, such as stronger robots.txt protocols, that specify permitted scraping behaviors and minimize overall reliance on CAPTCHAs. In a similar vein, we also recommend that researchers consider how user-specified UI patterns can impact authentication. For example, aggressive session timeouts, and lockouts, create substantial friction for users with limited mobility [45] and those using AT with longer input speeds. User-specified timeout lengths or fallback options, such as offering alternative authentication mechanisms before triggering a lockout, can make interactions more accessible to people with disabilities, but it is unclear if this would result in greater security and privacy harms. Of course, this would require active engagement between researchers, UX designers, and policy-makers. As explored in this work and previous [8, 13, 13, 23, 24, 33, 45, 53], both knowledge-based and other forms of authentication require wholesale assessment regarding its efficacy for disabled users. As researchers, we should look for new S&P paradigms that are inclusive to the needs of disabled users **and** their allies [3, 85]; as a prerequisite, this must involve disabled users throughout the research process as co-authors, designers, and advisors [89].

## 7 ACKNOWLEDGMENTS

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## 8 APPENDIX

### 8.1 Ethical Considerations

The interview and recruitment protocol was approved by our institution's Internal Review Board (IRB). However, we recognize that IRB approval is not enough to ensure ethical research. Our research

methodology was guided by the Menlo Report's [10] ethical principles. Stakeholders included BLV users, website operators, web developers, the Web Accessibility Initiative, accessibility audit tools, and security practitioners.

- **Respect for Persons:** Interview participants were informed of the study's purpose and scope during recruitment and before the interview. We also emphasized their right to skip uncomfortable questions without penalty or withdraw from the study at any point, no questions asked. Interviews were strictly limited to one hour to mitigate interviewee fatigue.
- **Beneficence: Benefits, Harms, and Mitigation:** This study potentially helps **improve** audit tools, security practitioners and web developers to create ADA-compliant websites that improve overall security, and make BLV users online experience less frustrating and accessible. Among the potential **harms** is that the automated scanning creates a traffic overhead for website operators and publishing negative results may impact the operators' reputation. To **mitigate** these harms, we kept the impact of our crawlers minimal by limiting interactions to two queries per webpage for automated audits, and a single follow-up visit for manual analysis. We also do not name and shame specific website operators in this work. To avoid interview participants revealing any personal data relevant for authentication, e.g., usernames, passwords, security questions.
- **Justice:** BLV users bore the burden of participating in the interviews. However, they are also the main benefactor of improved accessibility of account creation and authentication pages. All interviewees were compensated with \$25 Amazon gift cards for their time.
- **Respect for Law and Public Interest:** Our scanning approach based on Puppeteer and axe-core accessibility tools, did not circumvent any technical barriers to gain access, limiting itself to the use of the account creation and authentication pages. We respected websites' robots.txt file, rate limiting requests, and did not retry connections when they were blocked initially – avoiding any potential financial harm to the website operators. Since we do not republish visual depictions of the scanned websites, we also avoid potential problems with applicable copyright laws.

**Decision to conduct and publish the research.** In line with prior work [6, 34, 87], we weighed the minor overhead of automated scanning against the societal benefits of improving web accessibility and online security for disabled users. Publishing the research demonstrates the rigorous approach of the measurement approach, lends credibility to the results, and urges website operators, web developers, and security practitioners to adapt their current practices.

### 8.2 Open Science

The codebook, demographics, interview protocol, and analysis code can be found in OSF <sup>3</sup> repository. A subset of audit results is provided rather than the full dataset due to its significant size (exceeding 1GB), and to honor the privacy of the original dataset [6].

<sup>3</sup>[https://osf.io/hjb2m/overview?view\\_only=50acd8d7c49041b5990b3b215e9faa5a](https://osf.io/hjb2m/overview?view_only=50acd8d7c49041b5990b3b215e9faa5a)